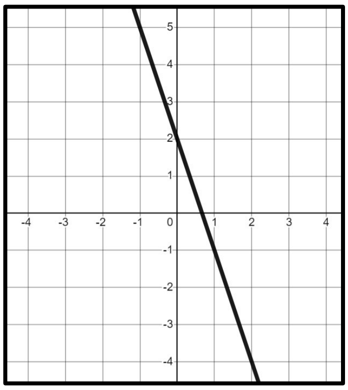
Negative rate of change from a graph

Task A: Calculating negative gradients from graphs

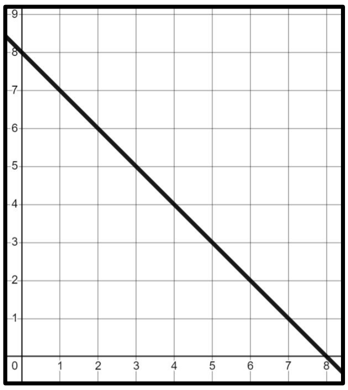
1) Find the gradient of each line.

a)



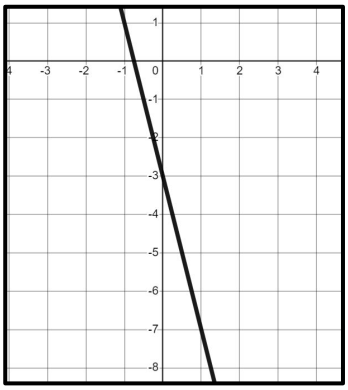
-3

b)



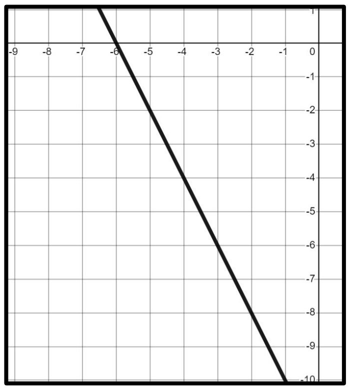
-1

c)



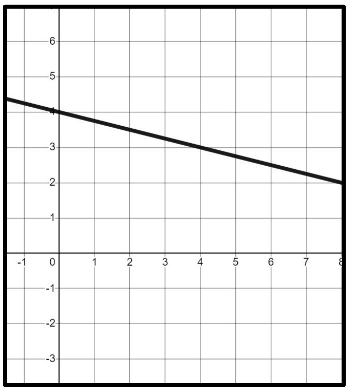
-4

d)



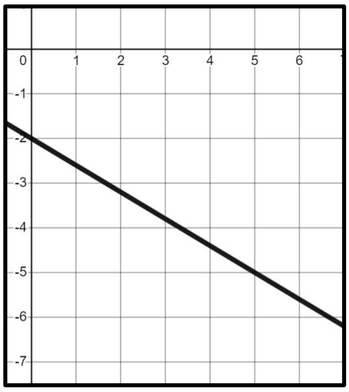
-2

e)



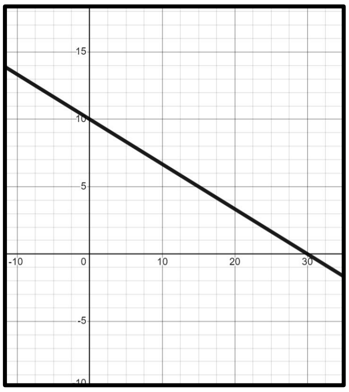
4	-1
1	$-\frac{1}{4}$

f)



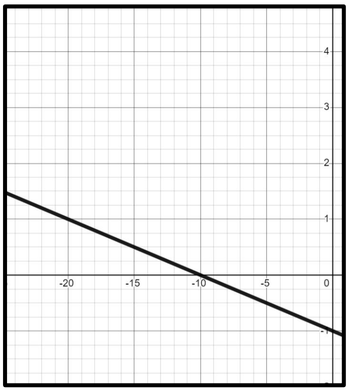
5	-3
1	$-\frac{3}{5}$

g)



30	-10
1	$-\frac{1}{3}$

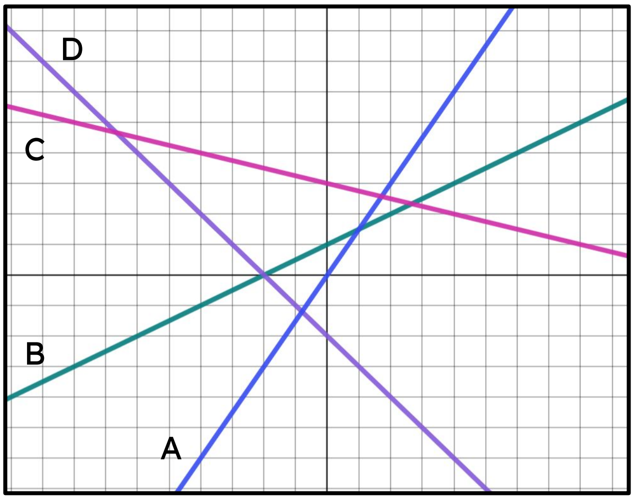
h)



10	-1
1	$-\frac{1}{10}$

Task B: Calculating gradient from any graph

1) Four lines have been drawn on a set of axes with unknown scale. Match each line to its gradient.



a) gradient = 1 *B (green line)*

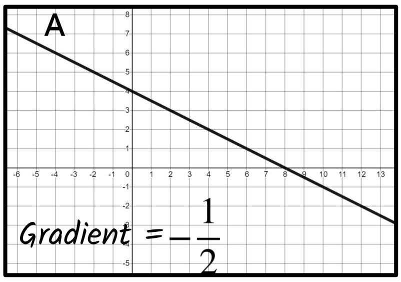
b) gradient = 3 *A (blue line)*

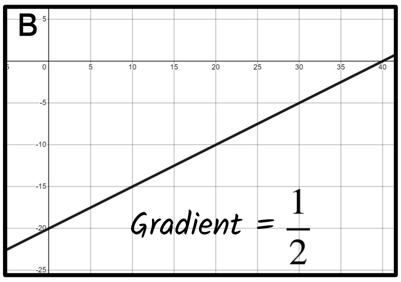
c) gradient = −2 *D (purple line)*

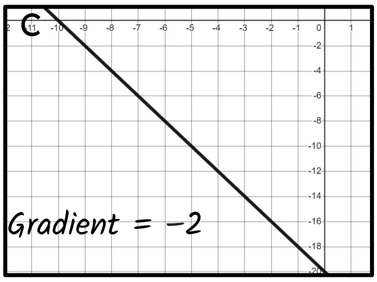
d) gradient = $-\frac{1}{2}$ *C (pink line)*

2) Which lines are parallel?

Calculate the gradients and match any parallel lines.

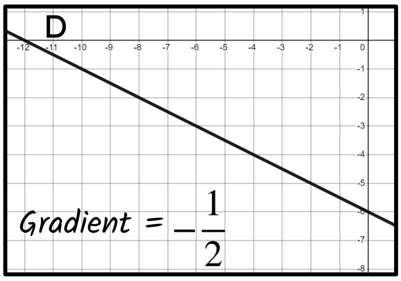


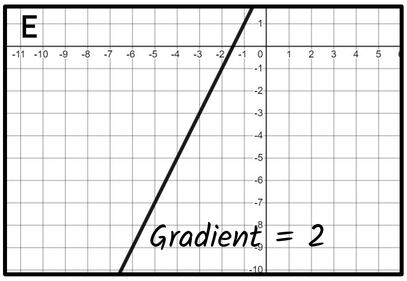


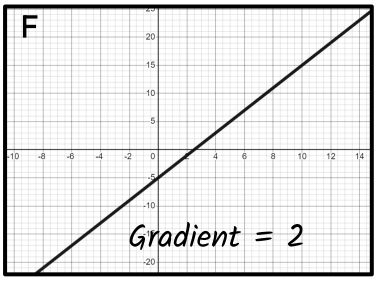


A and D are parallel

E and F are parallel

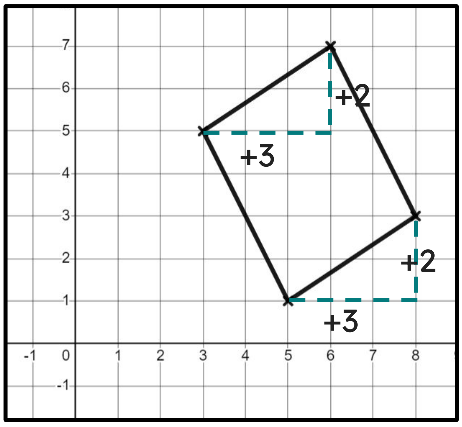




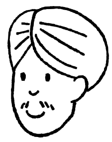


3) Andeep is trying to prove this shape is a parallelogram

Finish off his working out.



Opposite sides in a parallelogram are parallel. Therefore they must have the same gradient.



Andeep

The left and right sides are parallel…

Change in x	*Change in y*
3	*2*
1	$\frac{2}{3}$

The top and bottom sides have the same gradient of $\frac{2}{3}$ so they are parallel.